

Project Development Phase

Date	13 July 2026
Team ID	
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

Data Preprocessing Steps

Step 1: Import the Dataset

- Imported the Transformed_Housing_Data2.csv dataset into Tableau Public.
- Verified that all records and fields (Sale Price, Bedrooms, Bathrooms, Sqft Living, Sqft Basement, Condition, Year Built, Renovation info, Zipcode, Latitude/Longitude, etc.) were imported successfully.
- Confirmed that the dataset (~21,609 records) was ready for preprocessing.

Step 2: Check for Missing Values

- Examined the dataset for null or blank values.
- Verified that important fields such as Sale Price, Bedrooms, Bathrooms, Condition, and Renovation Year contained valid information.
- Missing values were handled appropriately to ensure accurate analysis.

Step 3: Remove Duplicate Records

- Checked the dataset for duplicate house records.
- Removed duplicate entries to prevent inaccurate counts and incorrect business insights.
- Ensured that every house sale was represented only once.

Step 4: Validate Data Types

Ensured that each attribute had the correct data type:

- Sale Price → Number
- Bedrooms / Bathrooms / Floors → Number
- Sqft Living / Sqft Basement → Number
- Condition → Categorical
- Year Built / Year Renovated → Date/Number
- Zipcode → Geographic/String
- Latitude / Longitude → Geographic/Number

Step 5: Verify Data Consistency

- Checked for invalid or unrealistic values such as negative sale prices, zero bedrooms/bathrooms where not applicable, or impossible renovation years (e.g., renovation year earlier than year built).
- Standardized condition ratings and zipcode formatting.
- Ensured consistency across all categorical and numerical fields before visualization.

Step 6: Prepare Calculated Fields

Created calculated fields required for dashboard analysis, including:

- Years Since Renovation
- House Age
- Average Sale Price
- Basement Area (from Sqft Basement)
- House Age Grouping (for bedroom/bathroom/floor analysis)
- KPI Measures for Dashboard (Count, Average Sale Price, Basement Area)

Step 7: Final Clean Dataset

- Verified that the cleaned dataset was complete and suitable for analysis.
- Confirmed that all preprocessing activities were completed successfully.
- The dataset was finalized for dashboard creation and business intelligence visualizations in Tableau Public.

Business Questions with Visualization

S.No.	Business Question	Visualization
1.	How many house sale records are available for analysis after data preprocessing?	Total Count (Text Table / KPI)
2.	What is the average sale price of houses in the dataset?	Average Sale Price (Text Table / KPI)
3.	What is the average basement area across houses?	Area from Basement (Text Table / KPI)
4.	How are total house sales distributed based on years since renovation?	Total Sales by Years Since Renovation (Histogram)
5.	What is the distribution of house age based on renovation status?	Distribution of House Age by Renovation Status (Pie Chart)
6.	How does house age distribution vary across bedrooms, bathrooms, and floors?	House Age Distribution by Bathrooms/Bedrooms/Floors (Bar Chart)

Visualizations

Figure 1: Text Table – Count / Average Sale Price / Basement Area

Area of the House from Basement (in Sqft)	38,643,798
Count of Transformed_Housing_Data2.csv	21,609
Avg. Sale Price	511,619

Figure 2: Total Sales by Years Since Renovation (Histogram)

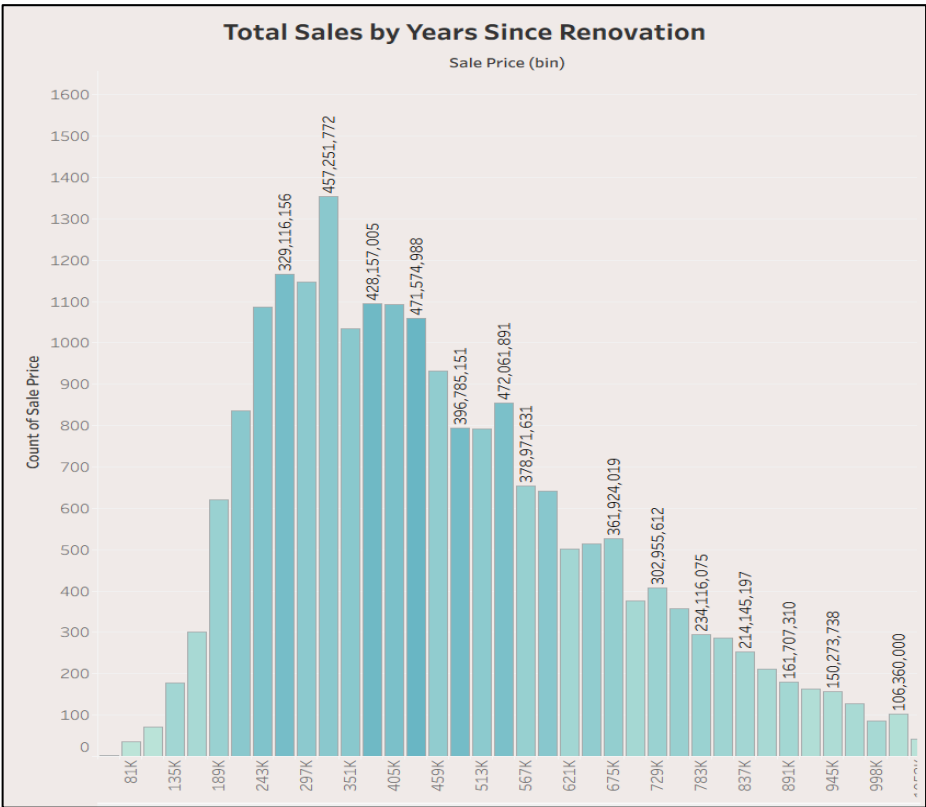


Figure 3: Distribution of House Age by Renovation Status (Pie Chart)

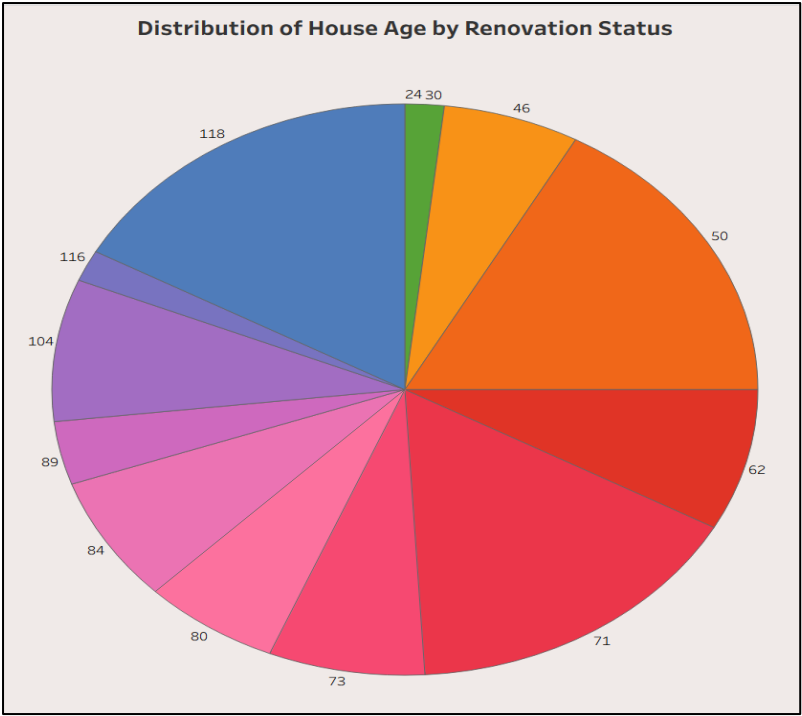


Figure 4: House Age Distribution by Bathrooms/Bedrooms/Floors (Bar Chart)

